University of Prompts & Logic

The Philosophy: "Literacy First, Velocity Second"

The University of Prompts & Logic was founded on a singular observation: **You cannot effectively direct an AI to build something if you do not understand the mechanics of how it is built.**

While other institutions rush to teach "Prompt Engineering" as an isolated skill, we treat AI as an acceleration layer on top of foundational competence. Our students do not just learn to generate; they learn to architect, debug, and refine.

The 12-Month Progression

Our academic year is strictly divided into two distinct phases to ensure deep competency:

- **Phase 1: The Manual Foundation (Months 1-6)**
    - Students are discouraged from using generative through these parts to ensure the fundamental Logic is learnt.
    - *Goal:* To understand the constraints, logic, and "first principles" of the work environment.
- **Phase 2: The AI Acceleration (Months 7-12)**
    - Students are encouraged to use the full suite of AI tools.
    - The focus shifts from "how to think about solving" to "how to get it done fast."
    - *Goal:* To use the knowledge from Phase 1 to spot AI hallucinations, fix errors, and chain workflows for exponential speed.

The "P&L" Methodology

The University operates on a strict weekly rhythm designed to test both technical understanding and adaptive reasoning.

- **Prompts (The "How"):** The ability to communicate intent to a machine.
- **Logic (The "Why"):** The ability to troubleshoot when the machine fails.

The Weekly Cycle:

1. **Monday (Input):** Video Lectures introducing concepts.

2. **Wednesday (Pressure Test):** The **Prompts & Logic (P&L) Quiz**. A specialized assessment that tests diagnostics (identifying hardware bottlenecks, code errors, or visual artifacts) and prompt syntax.

3. **Friday (Output):** Project Sprints where theory is applied to the year-long Capstone Project.

The Faculties

1. Faculty of Media

- **Mission:** Transforming artists into "Hybrid Creators."
- **Journey:** From understanding bitrates, keyframes, and color theory to mastering AI video interpolation, generative voice synthesis, and automated content scaling.
- **Outcome:** Graduates can produce a full multimedia brand experience single-handedly.

2. Faculty of Business

- **Mission:** Transforming managers into "System Architects."
- **Journey:** From manual data entry, writing emails, and setting calendar appointments to building autonomous agents, self-healing customer support bots, and predictive financial models.
- **Outcome:** Graduates can build an "Autonomous Enterprise" that runs with minimal human intervention.

3. Faculty of Coding

- **Mission:** Transforming coders into "Software Conductors."
- **Journey:** From fighting with terminal commands and syntax errors to directing AI to generate boilerplate, write documentation, and refactor legacy codebases across multiple platforms.
- **Outcome:** Graduates can deploy complex, secure, multi-platform ecosystems (Web, Mobile, Game) at the speed of a full development team.

Program Structure: 52 Weeks (4 Quarters)

- **Phase 1 (Weeks 1-24):** Manual Foundations & Environment Navigation.
- **Phase 2 (Weeks 25-52):** AI Integration & Workflow Mastery.

Faculty of Media

Phase 1: Foundations (Manual Mastery)

Month 1: The Creator's Rig & Audio Fundamentals

- **Week 1: Hardware & Specs**
  - *Topic:* CPU vs GPU for rendering, RAM requirements, SSD vs HDD.
  - *Wed P&L Quiz:* **Hardware Logic.** Given a specific bottleneck (e.g., "Video playback is stuttering but render speed is fine"), identify which component needs upgrading.
- **Week 2: File Management & Formats**
  - *Topic:* Lossless vs Lossy (WAV/FLAC vs MP3), Container types (PNG, JPEG, MOV, MP4, MKV) and converting between them.
  - *Wed P&L Quiz:* **Format Logic.** Scenario-based questions (e.g., "Client needs a transparent video background; which format do you choose?").
- **Week 3: The Audio Interface**
  - *Topic:* Audio input, Audio Repair, Audio replacement.
  - *Wed P&L Quiz:* **Signal Flow Logic.** Trace the path of sound from microphone to computer, then remove the noise and echo, then change the voice.
- **Week 4: Screen Recording & Extraction**
  - *Topic:* In built recording and capture, extracting audio from video files, isolating tracks.
  - *Wed P&L Quiz:* **Troubleshooting.** Identify why a screen recording has a black screen or desynced audio based on settings screenshots.

Month 2: Music Generation Software (Free vs Paid vs Licensed)

- **Week 5: DAW Interface Navigation**
  - *Topic:* Tracks, Timeline, Mixer, VST plugins.
  - *Wed P&L Quiz:* **UI Navigation.** "Click where the master volume fader is" (Hotspot quiz) or identifying icons for mute/solo.
- **Week 6: Rhythm & Structure**

    - *Topic:* Bars, Beats, Time Signatures, Intro/Verse/Chorus structure.
    - *Wed P&L Quiz:* **Structural Logic.** Listen to a clip and identify the "1" count or drag-and-drop song sections into the correct order.
- **Week 7: Splicing & Fades**
    - *Topic:* Cutting audio, crossfades, snapping to grid.
    - *Wed P&L Quiz:* **Editing Logic.** "Which crossfade type (Linear vs Logarithmic) prevents a sudden volume drop?"
- **Week 8: Intro to Generative Audio (Suno)**
    - *Topic:* The Suno Interface, Tags vs Lyrics, Extending tracks.
    - *Wed P&L Quiz:* **Tag Logic.** Match the music genre to the correct style tags (e.g., "Lo-fi" vs "Synthwave").

Month 3: Visuals & Image Composition

- **Week 9: Resolution & Aspect Ratios**
    - *Topic:* DPI/PPI, 16:9 vs 9:16, Upscaling artifacts, cropping.
    - *Wed P&L Quiz:* **Dimension Logic.** Calculate the pixel dimensions for a 4K 16:9 image vs a 1080p vertical story.
- **Week 10: Composition Rules**
    - *Topic:* Rule of Thirds, Leading Lines, Golden Ratio.
    - *Wed P&L Quiz:* **Visual Analysis.** Identify which rule of composition is being used in a provided image.
- **Week 11: Layers & Masking**
    - *Topic:* Photoshop/GIMP basics, Alpha channels, Blending modes.
    - *Wed P&L Quiz:* **Layer Logic.** "If Layer A is set to 'Multiply' over Layer B, what happens to the white pixels?"
- **Week 12: Intro to Image Gen (Midjourney/DALL-E)**
    - *Topic:* Parameters (--ar, --v), Seeds, Weights.
    - *Wed P&L Quiz:* **Parameter Logic.** Identify what error a specific parameter caused (e.g., "Why is this image square when I wanted landscape?").

Month 4: Video Production Environment

- **Week 13: Timelines & Frame Rates**
  - *Topic:* 24fps vs 30fps vs 60fps, Drop frame timecode.
  - *Wed P&L Quiz:* **Time Logic.** "If you slow down 24fps footage by 50%, why does it look choppy?"
- **Week 14: Cutting & Pacing**
  - *Topic:* J-cuts, L-cuts, Jump cuts, Matching action.
  - *Wed P&L Quiz:* **Edit Logic.** Identify a J-cut from a visual timeline representation.
- **Week 15: Keyframing Basics**
  - *Topic:* Interpolation (Linear vs Bezier), Ease-in/Ease-out.
  - *Wed P&L Quiz:* **Motion Logic.** Predict the movement of an object based on a graph curve (Speed vs Time).
- **Week 16: Exporting & Codecs**
  - *Topic:* H.264 vs H.265, Bitrate settings.
  - *Wed P&L Quiz:* **Compression Logic.** "Why is this exported file 10GB? Spot the wrong setting."

Month 5: Animation & Motion Basics

- **Week 17: Principles of Animation**
  - *Topic:* Squash and Stretch, Anticipation. Footage from screen recording code.
  - *Wed P&L Quiz:* **Physics Logic.** Identify the missing frame in a bouncing ball sequence.
- **Week 18: Prompting for Motion**
  - *Topic:* Camera movements (Pan, Tilt, Zoom, Truck).
  - *Wed P&L Quiz:* **Camera Logic.** Match the video clip to the correct camera term.
- **Week 19: Interpolation (AI Video)**
  - *Topic:* How AI "dreams" between frames.

    - *Wed P&L Quiz:* **Artifact Spotting.** Identify where the AI morphing failed in a sequence.

- **Week 20: Looping Fundamentals**
    - *Topic:* Creating seamless loops, boomerang effects.
    - *Wed P&L Quiz:* **Loop Logic.** Find the "seam" or error in a looping texture.

Month 6: Integration & Assembly

- **Week 21: Syncing Audio to Video**
    - *Topic:* Waveform alignment, visual cues.
    - *Wed P&L Quiz:* **Sync Logic.** "Is the audio ahead or behind the video in this clip?" (Test reaction/perception).
- **Week 22: Audio Mixing for Video**
    - *Topic:* Ducking, EQ for speech vs music.
    - *Wed P&L Quiz:* **Frequency Logic.** Identify which frequency range makes speech sound "muddy."
- **Week 23: Color Correction Basics**
    - *Topic:* White balance, Saturation, Contrast.
    - *Wed P&L Quiz:* **Color Logic.** "The image looks too blue; which color slider do you adjust to fix it?"
- **Week 24: The Rough Cut (Alpha)**
    - *Topic:* Review workflows, Version control.
    - *Wed P&L Quiz:* **Workflow Logic.** Organize a messy file folder structure into the correct hierarchy.

Phase 2: AI Mastery (Workflow & Speed)

Month 7: Advanced Prompting Architectures

- **Week 25: The Anatomy of a Prompt**
    - *Topic:* Subject + Medium + Style + Lighting + Color + Vibe. Economics and Limitations.

    - *Wed P&L Quiz:* **Prompt Debugging.** "Here is a bad prompt and the result. Fix the specific word causing the issue. when to start over in a new session?"

- **Week 26: Iterative Prompting**
    - *Topic:* Rerolling, Variations, Strong vs Weak variations.
    - *Wed P&L Quiz:* **Iteration Logic.** "Which variation setting (High vs Low) would you use to fix a hand without changing the face?"
- **Week 27: Negative Prompting**
    - *Topic:* Removing unwanted elements (--no parameters).
    - *Wed P&L Quiz:* **Negative Logic.** "The image has text overlays. What command removes them?"
- **Week 28: Image-to-Image Styles**
    - *Topic:* Using reference images, Style weights.
    - *Wed P&L Quiz:* **Weight Logic.** Predict the output based on the image weight setting (IW).

Month 8: AI Video Workflows

- **Week 29: Consistency Characters**
    - *Topic:* Seed locking, Character sheets in video.
    - *Wed P&L Quiz:* **Consistency Logic.** Identify which of 4 clips has a "character drift" error.
- **Week 30: AI In-Painting for Video**
    - *Topic:* Changing background elements in moving footage.
    - *Wed P&L Quiz:* **Masking Logic.** Select the correct area to mask for a specific effect.
- **Week 31: Frame Interpolation**
    - *Topic:* Increasing frame rate with AI (Slow motion).
    - *Wed P&L Quiz:* **Motion Artifacts.** Identify "ghosting" artifacts in slow-mo footage.
- **Week 32: Lip Syncing AI**

    - *Topic:* Syncing audio to AI avatars (HeyGen, SyncLabs).
    - *Wed P&L Quiz:* **Lip Sync Logic.** "Why is the mouth movement not matching the phoneme 'B'?"

Month 9: Audio Synthesis & Voice

- **Week 33: Voice Cloning Tech**
    - *Topic:* ElevenLabs parameters (Stability, Similarity).
    - *Wed P&L Quiz:* **Parameter Logic.** "The voice sounds too emotional and cracked. Which slider do you lower?"
- **Week 34: Text-to-Song Prompting**
    - *Topic:* Metatags [Verse], [Chorus], [Solo].
    - *Wed P&L Quiz:* **Structure Prompting.** Write the correct metatags to force a guitar solo after the second chorus.
- **Week 35: Stem Splitting**
    - *Topic:* AI separation of vocals and instrumentals.
    - *Wed P&L Quiz:* **Audio Isolation.** Identify bleed-through artifacts in an extracted vocal track.
- **Week 36: Audio Cleanup**
    - *Topic:* AI De-noise and De-reverb.
    - *Wed P&L Quiz:* **Spectral Logic.** Identify the visual representation of "noise" in a spectral graph.

Month 10: Troubleshooting & Editing

- **Week 37: Hallucination Spotting**
    - *Topic:* Hands, Text, Physics errors.
    - *Wed P&L Quiz:* **"Where's Waldo" Error Spotting.** Find the 5 AI errors in a complex image.
- **Week 38: Out-Painting (Expanding Canvas)**
    - *Topic:* Changing aspect ratios post-generation.

    - *Wed P&L Quiz:* **Perspective Logic.** "If we expand this image left, what *should* be there based on the perspective lines?"

- **Week 39: Upscaling Workflows**
    - *Topic:* Creative Upscale vs Faithful Upscale.
    - *Wed P&L Quiz:* **Upscale Logic.** "The upscale added details that weren't there. Which mode was used?"

- **Week 40: Fixing AI Hands/Faces**
    - *Topic:* Photoshop Generative Fill for specific fixes.
    - *Wed P&L Quiz:* **Selection Logic.** "Draw the optimal selection shape to fix this distorted hand."

Month 11: Speed Workflows

- **Week 41: Batch Generation**
    - *Topic:* Generating assets in bulk (Permutations).
    - *Wed P&L Quiz:* **Permutation Logic.** Calculate how many images {Red, Blue} {Cat, Dog} --v 5 will generate.

- **Week 42: Automation Scripts**
    - *Topic:* Automating file naming and sorting.
    - *Wed P&L Quiz:* **Sorting Logic.** Create a logical rule for organizing 1000 files by date and type.

- **Week 43: Content Repurposing**
    - *Topic:* Turning 1 video into 3 Shorts/Reels.
    - *Wed P&L Quiz:* **Cropping Logic.** Choose the best crop area for vertical video from a horizontal shot.

- **Week 44: Speed Editing**
    - *Topic:* Keyboard shortcuts and AI cut detection.
    - *Wed P&L Quiz:* **Shortcut Memory.** Timed quiz on essential keyboard shortcuts.

Month 12: Final Assembly

- **Week 45: Portfolio Packaging**
  - *Topic:* Behance/ArtStation layouts.
  - *Wed P&L Quiz:* **Curation Logic.** Select the strongest 3 images from a set of 10.
- **Week 46: Presentation Skills**
  - *Topic:* Explaining your AI workflow to clients.
  - *Wed P&L Quiz:* **Client Comm Logic.** "How do you explain to a client why you can't just 'change the angle' of a generated image?"
- **Week 47: Final Polish**
  - *Topic:* Final render checks.
  - *Wed P&L Quiz:* **QC Logic.** Final Quality Control checklist.
- **Week 48: Graduation & Demo Day**
  - *Topic:* Live Presentation.
  - *Wed P&L Quiz:* **Reflection.** Self-assessment logic.

Faculty of Business

Phase 1: Foundations (Business Logic)

Month 1: Market Entry & Business Security

- **Week 1: Niche Selection & Research**
  - *Topic:* Identifying viable markets, Competitor analysis basics.
  - *Wed P&L Quiz:* **Market Logic.** "Given these 3 competitor prices, find the gap in the market."
- **Week 2: Business Models & Strategy**
  - *Topic:* B2B vs B2C logic, ROI, CAC (Customer Acquisition Cost).
  - *Wed P&L Quiz:* **Strategy Logic.** "Is this service better suited for a subscription or a one-time fee model?"
- **Week 3: Digital Security Fundamentals**
  - *Topic:* VPNs, 2FA, Password Managers, Phishing recognition.
  - *Wed P&L Quiz:* **Security Logic.** "Which of these emails is a phishing attempt?"
- **Week 4: Business Computing Specs**
  - *Topic:* Hardware for multitasking, multiple monitors, cloud storage vs local.
  - *Wed P&L Quiz:* **Tech Logic.** "Which laptop spec is most critical for running 50 browser tabs?"

Month 2: Data & Form Logic

- **Week 5: Form Logic Fundamentals**
  - *Topic:* Conditional Logic (If/Then), Branching questions.
  - *Wed P&L Quiz:* **Conditional Logic.** "If the user selects 'No' to question 2, which question should they skip to?"
- **Week 6: Database Basics**
  - *Topic:* Relational vs Flat databases, CSV files, Data types.

    - *Wed P&L Quiz:* **Data Structure Logic.** "Match the data (Phone Number) to the correct field type (Integer vs String)."

- **Week 7: Privacy & Compliance**
    - *Topic:* GDPR basics, Data handling, Privacy policies.
    - *Wed P&L Quiz:* **Compliance Logic.** "Identify the violation in this data collection form."

- **Week 8: Creating Intake Forms**
    - *Topic:* Building the Client Intake flow (Fillout/Typeform).
    - *Wed P&L Quiz:* **UX Logic.** "Why did the user abandon the form at Step 3?"

Month 3: Communication Infrastructure

- **Week 9: Email Protocols**
    - *Topic:* SMTP, IMAP, DNS records for email (DKIM/SPF).
    - *Wed P&L Quiz:* **Protocol Logic.** "Why did this email go to spam? Check the DNS settings."

- **Week 10: Writing Copy**
    - *Topic:* Hooks, CTAs (Call to Action), Tone of voice.
    - *Wed P&L Quiz:* **Copy Logic.** "Identify the weak 'Call to Action' in this email."

- **Week 11: VOIP & Phone Systems**
    - *Topic:* Virtual numbers, Call routing, Voicemail logic.
    - *Wed P&L Quiz:* **Routing Logic.** "Design a call flow: Press 1 for Sales, Press 2 for Support."

- **Week 12: CRM Fundamentals**
    - *Topic:* Manual CRM entry, Lead statuses, Pipelines.
    - *Wed P&L Quiz:* **Pipeline Logic.** "Where does this lead belong? (Cold, Warm, Hot)."

Month 4: Scheduling & Logistics

- **Week 13: Calendar Architecture**

    - *Topic:* Calendar APIs, Syncing multiple calendars.
    - *Wed P&L Quiz:* **Sync Logic.** "Why is the calendar showing 'Busy' when I'm free?"
- **Week 14: Time Zone Logistics**
    - *Topic:* UTC, Time zone math, DST handling.
    - *Wed P&L Quiz:* **Scheduling Logic.** "If it's 3PM EST, what time is it in Tokyo? Will this meeting conflict?"
- **Week 15: Conflict Resolution**
    - *Topic:* Manual scheduling buffers, Overbooking prevention.
    - *Wed P&L Quiz:* **Logic Puzzle.** "Fit these 4 meetings into this 2-hour window."
- **Week 16: Booking Page Setup**
    - *Topic:* Setting up Calendly/Cal.com, Integration testing.
    - *Wed P&L Quiz:* **Settings Logic.** "User can't book on Fridays. Find the setting error."

Month 5: Financial Systems

- **Week 17: Payment Gateways**
    - *Topic:* Stripe/PayPal setup, API keys (Test vs Live).
    - *Wed P&L Quiz:* **Gateway Logic.** "The payment failed with Error 402. What does this code usually mean?"
- **Week 18: Invoicing Logic**
    - *Topic:* Invoice terms (Net30), Line items, VAT/Tax.
    - *Wed P&L Quiz:* **Billing Logic.** "Calculate the total tax for this invoice based on the region."
- **Week 19: Refunds & Disputes**
    - *Topic:* Handling chargebacks, Refund policies.
    - *Wed P&L Quiz:* **Policy Logic.** "Is this customer eligible for a refund based on the policy text?"
- **Week 20: Financial Dashboarding**

    - *Topic:* Reading dashboards, MRR, Churn rates.
    - *Wed P&L Quiz:* **Metric Logic.** "Churn increased by 5%. Look at the data and guess why."

Month 6: Customer Support Infrastructure

- **Week 21: Empathy & Psychology**
    - *Topic:* De-escalation techniques, Reading customer sentiment.
    - *Wed P&L Quiz:* **Sentiment Analysis.** "Is this ticket Angry, Frustrated, or Confused?"
- **Week 22: Tagging & Categorization**
    - *Topic:* Ticket routing, Priority levels.
    - *Wed P&L Quiz:* **Categorization Logic.** "Tag these 5 support tickets with the correct category (Billing, Tech, Feature Request)."
- **Week 23: Standard Operating Procedures (SOPs)**
    - *Topic:* Writing clear instructions, Process mapping.
    - *Wed P&L Quiz:* **Process Logic.** "Reorder these steps to make the SOP efficient."
- **Week 24: Knowledge Base Construction**
    - *Topic:* Structuring help articles, Searchability.
    - *Wed P&L Quiz:* **Information Architecture.** "Where should the 'How to Reset Password' article live in the menu?"

Phase 2: AI Automation (Scale)

Month 7: AI Communication

- **Week 25: Prompting for Tone**
    - *Topic:* Adjusting AI persona (Professional vs Casual).
    - *Wed P&L Quiz:* **Tone Logic.** "This email sounds too aggressive. Select the prompt modifier that softens it."
- **Week 26: AI Rewriting & Editing**
    - *Topic:* Using AI to refine copy, Expanding bullets to paragraphs.

    - *Wed P&L Quiz:* **Editing Logic.** "Select the AI output that best summarizes this meeting transcript."

- **Week 27: AI Translation**
    - *Topic:* Localization, Handling idioms with AI.
    - *Wed P&L Quiz:* **Localization Logic.** "The AI translated this literally. Spot the cultural error."

- **Week 28: Automated Summaries**
    - *Topic:* Extracting action items from text.
    - *Wed P&L Quiz:* **Extraction Logic.** "Identify the 3 action items the AI missed in this summary."

Month 8: Chatbot Engineering

- **Week 29: Chatbot Architecture**
    - *Topic:* Decision trees vs LLMs, Knowledge base linking.
    - *Wed P&L Quiz:* **Flow Logic.** "Draw the path the bot takes when the user says 'Refund'."

- **Week 30: Training Data formatting**
    - *Topic:* Cleaning data for RAG (Retrieval-Augmented Generation).
    - *Wed P&L Quiz:* **Data Prep.** "Which of these documents will confuse the bot?"

- **Week 31: Temperature & Parameters**
    - *Topic:* Hallucination control, Creativity settings.
    - *Wed P&L Quiz:* **Parameter Logic.** "The bot made up a fake product. Was the Temperature too high or low?"

- **Week 32: Security & Injection Defense**
    - *Topic:* Prompt injection attacks, Guardrails.
    - *Wed P&L Quiz:* **Safety Logic.** "User tried to trick the bot into giving a discount. Did the bot fail?"

Month 9: Advanced Automation (Zapier/Make)

- **Week 33: Triggers & Actions**
  - *Topic:* Webhooks, Polling, Instant triggers.
  - *Wed P&L Quiz:* **Trigger Logic.** "Why didn't the automation fire when the form was submitted?"
- **Week 34: API Keys & Auth**
  - *Topic:* Connecting services, Secret management.
  - *Wed P&L Quiz:* **Connection Logic.** "The connection failed. Identify the invalid API Key format."
- **Week 35: Data Payloads**
  - *Topic:* JSON in automation, Mapping fields.
  - *Wed P&L Quiz:* **Mapping Logic.** "You mapped 'First Name' to the 'Email' field. Spot the error."
- **Week 36: Error Handling**
  - *Topic:* Filters, Routers, Error paths.
  - *Wed P&L Quiz:* **Workflow Logic.** "The automation failed at Step 3. Look at the data payload and find the null value."

Month 10: Data Analysis with AI

- **Week 37: AI Data Cleaning**
  - *Topic:* Formatting messy CSVs with prompts.
  - *Wed P&L Quiz:* **Cleanup Logic.** "Write a prompt to remove all duplicate emails from this list."
- **Week 38: Trend Analysis**
  - *Topic:* Spotting patterns in large datasets.
  - *Wed P&L Quiz:* **Data Logic.** "The AI says sales doubled, but the graph is flat. Identify the hallucination."
- **Week 39: Chart Generation**
  - *Topic:* Visualizing data with AI tools.

    - *Wed P&L Quiz:* **Visualization Logic.** "Which chart type (Bar vs Line) is best for showing growth over time?"

- **Week 40: Predictive Concepts**
    - *Topic:* Forecasting basics using historical data.
    - *Wed P&L Quiz:* **Prediction Logic.** "Based on this trend, predict next month's revenue."

Month 11: Complex Systems

- **Week 41: Circular Workflows**
    - *Topic:* Loops, Iterators in automation.
    - *Wed P&L Quiz:* **Loop Logic.** "Design a workflow that checks for new emails every hour."
- **Week 42: Multi-step Agents**
    - *Topic:* AI agents that browse, scrape, and report.
    - *Wed P&L Quiz:* **Agent Logic.** "The agent got stuck in a loop. Identify the missing 'Stop' condition."
- **Week 43: System Stress Testing**
    - *Topic:* Load testing, Rate limits.
    - *Wed P&L Quiz:* **System Logic.** "Design a workflow that retries a failed payment email 3 times before giving up."
- **Week 44: Redundancy**
    - *Topic:* Backups, Fallback automations.
    - *Wed P&L Quiz:* **Backup Logic.** "If OpenAI goes down, which fallback service should triggers?"

Month 12: Optimization & Launch

- **Week 45: Cost Optimization**
    - *Topic:* Token usage, API costs.
    - *Wed P&L Quiz:* **Optimization Logic.** "This workflow costs $0.50 per run. How do we optimize the token usage to get it to $0.10?"

- **Week 46: Documentation Generation**
  - *Topic:* Using AI to write technical docs.
  - *Wed P&L Quiz:* **Doc Logic.** "Does this AI-generated manual match the actual workflow?"
- **Week 47: Launch Prep**
  - *Topic:* Final system checks, User acceptance testing.
  - *Wed P&L Quiz:* **Launch Logic.** "Create a 'Go/No-Go' checklist for launch."
- **Week 48: Graduation**
  - *Topic:* Final Demo Day.
  - *Wed P&L Quiz:* **Reflection.** "How would you improve this system v2.0?"

Faculty of Coding

Phase 1: Foundations (Dev Environment)

Month 1: The Terminal & Environment

- **Week 1: Hardware & OS**
  - *Topic:* Specs for compiling, Mac vs Windows vs Linux for devs.
  - *Wed P&L Quiz:* **Hardware Logic.** "Which OS is native for iOS development?"
- **Week 2: VS Code Setup**
  - *Topic:* IDEs, Extensions (Prettier, ESLint), Themes.
  - *Wed P&L Quiz:* **IDE Logic.** "Which extension would fix this messy indentation automatically?"
- **Week 3: Terminal Basics**
  - *Topic:* Navigation (cd, ls), File manipulation (mkdir, touch).
  - *Wed P&L Quiz:* **Terminal Logic.** "Navigate around the terminal till you find the file 'P&L.md'?"
- **Week 4: Version Control (Git)**
  - *Topic:* Git init, commit, push, pull.
  - *Wed P&L Quiz:* **Git Logic.** "You tried to push but got a 'conflict' error. Why?"

Month 2: Web Structure

- **Week 5: How the Web Works**
  - *Topic:* DNS, IP addresses, Servers, Hosting.
  - *Wed P&L Quiz:* **Network Logic.** "Type the URL into the browser. What happens next?"
- **Week 6: HTML Structure**
  - *Topic:* Semantic HTML, Tags, Attributes.
  - *Wed P&L Quiz:* **Structure Logic.** "Spot the unclosed tag in this HTML snippet."

- **Week 7: CSS Fundamentals**
  - *Topic:* Box Model (Margin/Padding), Flexbox.
  - *Wed P&L Quiz:* **CSS Logic.** "The text is overflowing the box. Which CSS property is missing?"
- **Week 8: DOM Manipulation**
  - *Topic:* JavaScript basics for selecting elements.
  - *Wed P&L Quiz:* **DOM Logic.** "Write the selector to find the button with id='submit'."

Month 3: Data & APIs

- **Week 9: JSON Structure**
  - *Topic:* Objects, Arrays, Key-Value pairs.
  - *Wed P&L Quiz:* **JSON Logic.** "Find the syntax error (missing comma) in this JSON snippet."
- **Week 10: API Endpoints**
  - *Topic:* GET vs POST, Status codes (200, 404, 500).
  - *Wed P&L Quiz:* **API Logic.** "You sent data but got a 404. What does that mean?"
- **Week 11: Authentication**
  - *Topic:* API Keys, Bearer Tokens, OAuth flow.
  - *Wed P&L Quiz:* **Auth Logic.** "Where do you put the API key? (Header vs Body)."
- **Week 12: Hashing & Security**
  - *Topic:* Encryption basics, Salting passwords.
  - *Wed P&L Quiz:* **Security Logic.** "Why is storing passwords in plain text bad?"

Month 4: 3D & Game Logic

- **Week 13: Engine Interfaces**
  - *Topic:* Unity/Roblox Studio UI, Scene view vs Game view.

    - *Wed P&L Quiz:* **Engine Logic.** "Switch the camera from Perspective to Orthographic view."

- **Week 14: Coordinate Systems**
    - *Topic:* X, Y, Z axis, World space vs Local space.
    - *Wed P&L Quiz:* **3D Logic.** "If the parent object moves +10 on X, where does the child object go?"

- **Week 15: Physics & Rigidbodies**
    - *Topic:* Gravity, Mass, Velocity.
    - *Wed P&L Quiz:* **Physics Logic.** "Why did the object fall through the floor? (Missing Collider)."

- **Week 16: Colliders & Triggers**
    - *Topic:* Box colliders, Mesh colliders, OnEnter events.
    - *Wed P&L Quiz:* **Event Logic.** "Trigger the sound only when the player enters the zone."

Month 5: Mobile & Responsive

- **Week 17: Mobile Constraints**
    - *Topic:* Screen sizes, Touch targets, Battery usage.
    - *Wed P&L Quiz:* **UX Logic.** "Is this button too small for a thumb tap?"

- **Week 18: Responsive Breakpoints**
    - *Topic:* Media queries, Fluid layouts.
    - *Wed P&L Quiz:* **Responsive Logic.** "On a mobile screen, the menu disappears. Check the Media Query breakpoint."

- **Week 19: Emulation**
    - *Topic:* Simulators vs Emulators vs Real Devices.
    - *Wed P&L Quiz:* **Testing Logic.** "The feature works on simulator but fails on device. Why?"

- **Week 20: App Store Guidelines**
    - *Topic:* Review process, Sandbox restrictions.

    - *Wed P&L Quiz:* **Policy Logic.** "Apple rejected the app. Find the guideline violation."

Month 6: Programming Logic

- **Week 21: Variables & Types**
    - *Topic:* Strings, Integers, Booleans, Scope.
    - *Wed P&L Quiz:* **Type Logic.** "What happens if you add '5' (string) + 5 (number)?"
- **Week 22: Control Flow**
    - *Topic:* If/Else, Switch statements.
    - *Wed P&L Quiz:* **Logic Flow.** "Trace the path of this function if x = 10."
- **Week 23: Loops**
    - *Topic:* For loops, While loops, Iteration.
    - *Wed P&L Quiz:* **Loop Logic.** "How many times will this for loop run before crashing?"
- **Week 24: Functions**
    - *Topic:* Arguments, Returns, Reusability.
    - *Wed P&L Quiz:* **Function Logic.** "Refactor this code into a reusable function."

Phase 2: AI-Assisted Dev (Architect)

Month 7: Prompt Engineering for Code

- **Week 25: Context Management**
    - *Topic:* Token limits, providing context files.
    - *Wed P&L Quiz:* **Context Logic.** "The AI forgot the code from the previous file. Why? (Token limit exceeded)."
- **Week 26: System Prompts**
    - *Topic:* Setting the AI persona ("Act as a Senior Dev").
    - *Wed P&L Quiz:* **Prompt Logic.** "Write a system prompt that forces the AI to use Python."

- **Week 27: Boilerplate Generation**
    - *Topic:* Generating project skeletons.
    - *Wed P&L Quiz:* **Scaffolding Logic.** "Generate a React folder structure."
- **Week 28: Pastebin & Sharing**
    - *Topic:* Formatting code for AI (Markdown blocks).
    - *Wed P&L Quiz:* **Formatting Logic.** "The AI misread the indentation. Fix the paste format."

Month 8: Debugging & Refactoring

- **Week 29: Error Logs**
    - *Topic:* Reading stack traces, identifying root causes.
    - *Wed P&L Quiz:* **Debug Logic.** "Here is a stack trace. Select the prompt that correctly asks AI to fix the specific line."
- **Week 30: AI Debugging Workflows**
    - *Topic:* Paste-to-Debug, Iterative fixing.
    - *Wed P&L Quiz:* **Fix Logic.** "The AI suggested 3 fixes. Which one breaks the least amount of other code?"
- **Week 31: Refactoring**
    - *Topic:* Cleaning up code, optimization prompts.
    - *Wed P&L Quiz:* **Clean Code Logic.** "Ask AI to rename these variables to be more descriptive."
- **Week 32: Commenting**
    - *Topic:* Auto-generating JSDoc/Docstrings.
    - *Wed P&L Quiz:* **Doc Logic.** "Does the comment match the function behavior?"

Month 9: Generative Logic

- **Week 33: NLP to Code**
    - *Topic:* Codex, Copilot, Ghostwriter.

    - *Wed P&L Quiz:* **NLP Logic.** "Translate this user requirement ('I want the door to open when I have the key') into an If/Then logic statement."

- **Week 34: Game Scripting**
    - *Topic:* Generating C# / Lua behaviors.
    - *Wed P&L Quiz:* **Script Logic.** "Generate a script to make the player jump."
- **Week 35: NPC Logic**
    - *Topic:* State machines generated by AI.
    - *Wed P&L Quiz:* **AI Behavior.** "The NPC is stuck walking into a wall. Debug the state machine."
- **Week 36: Complex Logic**
    - *Topic:* Algorithms, Sorting logic.
    - *Wed P&L Quiz:* **Algo Logic.** "Ask AI to write the most efficient sorting algorithm for this list."

Month 10: Translation & Platforms

- **Week 37: Framework Translation**
    - *Topic:* Converting React to React Native.
    - *Wed P&L Quiz:* **Translation Logic.** "The AI used a Web-only library for a Mobile app. Spot the invalid import."
- **Week 38: Platform Fixes**
    - *Topic:* Android vs iOS specific bugs.
    - *Wed P&L Quiz:* **Platform Logic.** "This permission works on Android but fails on iOS. Why?"
- **Week 39: Dependency Management**
    - *Topic:* npm, yarn, AI package suggestions.
    - *Wed P&L Quiz:* **Package Logic.** "The AI suggested a deprecated package. Find the modern alternative."
- **Week 40: Optimization**
    - *Topic:* Memory leaks, Performance profiling.

    - *Wed P&L Quiz:* **Perf Logic.** "Identify the memory leak in this loop."

Month 11: Security & Efficiency

- **Week 41: AI Red Teaming**
    - *Topic:* Asking AI to hack your own code.
    - *Wed P&L Quiz:* **Security Logic.** "The AI generated code that allows a user to delete the database. Spot the vulnerability."
- **Week 42: SQL Injection**
    - *Topic:* Sanitizing inputs with AI checks.
    - *Wed P&L Quiz:* **Injection Logic.** "Spot the un-sanitized input field."
- **Week 43: Big O Notation**
    - *Topic:* Efficiency, Complexity analysis.
    - *Wed P&L Quiz:* **Complexity Logic.** "Which of these 2 functions is O(n) vs O(n^2)?"
- **Week 44: Stress Testing**
    - *Topic:* Simulating high load.
    - *Wed P&L Quiz:* **Load Logic.** "The server crashed at 1000 users. Why?"

Month 12: Deployment

- **Week 45: ReadMe Generation**
    - *Topic:* Writing clear instructions for users.
    - *Wed P&L Quiz:* **Documentation Logic.** "Match the function to the correct AI-generated description."
- **Week 46: Release Notes**
    - *Topic:* Changelogs, Versioning (SemVer).
    - *Wed P&L Quiz:* **Version Logic.** "Is this update a Patch, Minor, or Major version?"
- **Week 47: Deployment Scripts**
    - *Topic:* CI/CD pipelines, Dockerfiles.

    - *Wed P&L Quiz:* **Deploy Logic.** "The build failed in the pipeline. Check the logs."
- **Week 48: Final Launch**
    - *Topic:* Live Demo.
    - *Wed P&L Quiz:* **Final Exam.** Comprehensive logic check.